

第八讲：网络模块和派发模块

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① 前情提要

② 协议

③ 网络模块

④ 派发模块

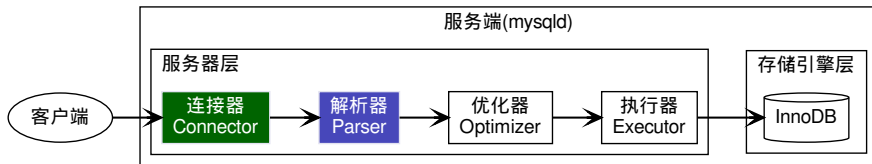


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前情提要



执行流程



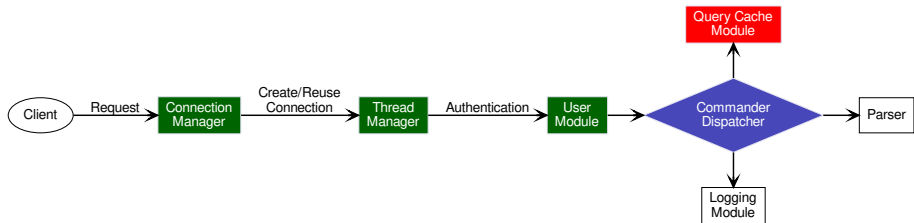
本节内容

- 连接器

- ▶ ☒ 连接管理器 Connection Manager
- ▶ ☒ 线程管理器 Thread Manager
- ▶ ☒ 用户模块 User Module

- 解析器

- ▶ ☐ 网络模块 Net Module
- ▶ ☐ 派发模块 Commander Dispatcher



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协议



客户端/服务器的协议

MySQL 的客户端支持以下数据交互协议：

- ① Connectors (Connector/C, Connector/J, and so forth)
- ② MySQL Proxy
- ③ Communication between master and slave replication servers

数据交互具体包含以下特性：

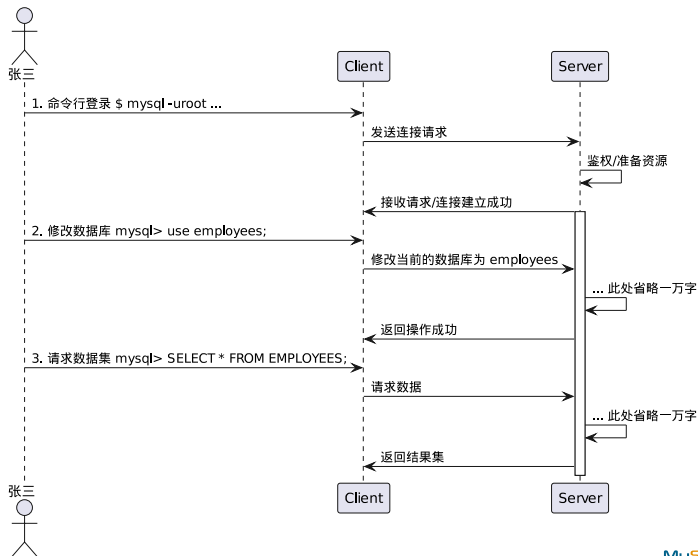
- ① 使用 SSL 进行透明数据加密 (Transparent encryption)
- ② 支持透明数据压缩 (Transparent compression)
- ③ 连接阶段¹ (Connection Phase)，处理兼容性和认证信息的数据交换
 - ▶ 在 `check_connection()` 函数中执行握手操作
- ④ 命令阶段² (Command Phase)，接收客户端命令然后执行
 - ▶ 在 `do_command()` 函数中执行用户命令

¹https://dev.mysql.com/doc/dev/mysql-server/8.0.37/page_protocol_connection_phase.html

²https://dev.mysql.com/doc/dev/mysql-server/8.0.37/page_protocol_command_phase.html



什么是命令



Command 枚举类型

- 客户端发送带序列号的包³
 - ▶ 序列号 sequence-id⁴，位于负载 (payload) 的第一个字节
 - ▶ 包 packet，负载后续传输的数据
- 所有命令定义见 `include/my_command.cc`，是 `enum enum_server_command` 枚举类型
- Text Protocol 文本协议
 - ▶ COM_QUERY
- Utility Commands 管理命令
 - ▶ COM_QUIT COM_INIT_DB COM_FIELD_LIST COM_REFRESH COM_STATISTICS
 - ▶ COM_PROCESS_INFO COM_PROCESS_KILL COM_DEBUG COM_PING
 - ▶ COM_CHANGE_USER COM_RESET_CONNECTION COM_SET_OPTION
- Prepared Statements 准备语句
 - ▶ COM_STMT_PREPARE COM_STMT_EXECUTE COM_STMT_FETCH COM_STMT_CLOSE
 - ▶ COM_STMT_RESET COM_STMT_SEND_LONG_DATA
- Stored Programs⁵
 - ▶ Multi-Resultset / Multi-Statement

³https://dev.mysql.com/doc/dev/mysql-server/8.0.37/page_protocol_command_phase.html

⁴https://dev.mysql.com/doc/dev/mysql-server/8.0.37/page_protocol_basic_packets.html

⁵https://dev.mysql.com/doc/dev/mysql-server/8.0.37/page_protocol_command_phase_sp.html



COM_QUERY 命令类型的负载示例⁶

Payload		
Type	Name	Description
int<1>	command	0x03: COM_QUERY
if CLIENT_QUERY_ATTRIBUTES is set {		
int<lenenc>	parameter_count	Number of parameters
int<lenenc>	parameter_set_count	Number of parameter sets. Currently always 1
if parameter_count > 0 {		
binary<var>	null_bitmap	NULL bitmap, length= (num_params + 7) / 8
int<1>	new_params_bind_flag	Always 1. Malformed packet error if not 1
if new_params_bind_flag, for each parameter {		
int<2>	param_type_and_flag	Parameter type (2 bytes). The MSB is reserved for unsigned flag
string<lenenc>	parameter name	String
}		
binary<var>	parameter_values	value of each parameter: Binary Protocol Value
}		
}		
string<EOF>	query	the text of the SQL query to execute

⁶https://dev.mysql.com/doc/dev/mysql-server/8.0.37/page_protocol_com_query.html



COM_DATA 联合体

- COM_DATA 定义见 `include/mysql/com_data.h`

```
112 union COM_DATA {  
113     COM_INIT_DB_DATA com_init_db;  
114     COM_REFRESH_DATA com_refresh;  
115     COM_KILL_DATA com_kill;  
116     COM_SET_OPTION_DATA com_set_option;  
117     COM_STMT_EXECUTE_DATA com_stmt_execute;  
118     COM_STMT_FETCH_DATA com_stmt_fetch;  
119     COM_STMT_SEND_LONG_DATA_DATA com_stmt_send_long_data;  
120     COM_STMT_PREPARE_DATA com_stmt_prepare;  
121     COM_STMT_CLOSE_DATA com_stmt_close;  
122     COM_STMT_RESET_DATA com_stmt_reset;  
123     COM_QUERY_DATA com_query;  
124     COM_FIELD_LIST_DATA com_field_list;  
125 };
```

- 它表示不同 Command 所需要关心的数据结构



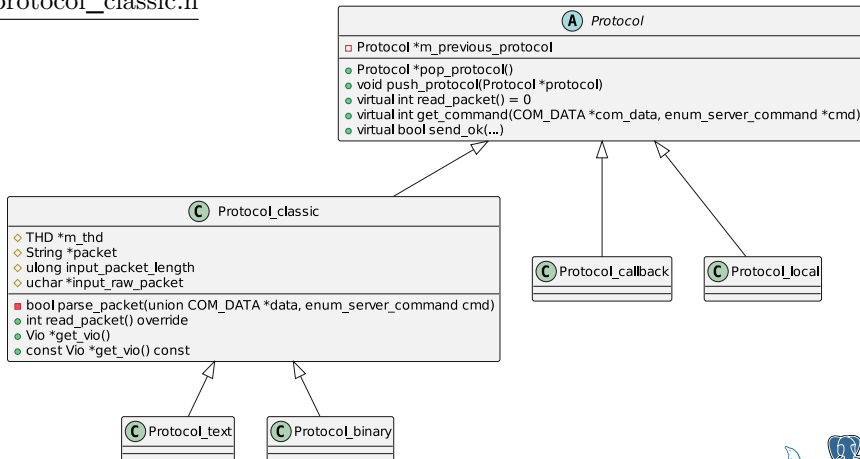
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网络模块



Protocol 协议类

- 抽象类 Protocol 实现见文件 `sql/protocol.h`，它实现了 MySQL 交互协议⁷
- 实现类 Protocol_classic / Protocol_text / Protocol_binary 见文件 `sql/protocol_classic.h`



⁷https://dev.mysql.com/doc/dev/mysql-server/8.0.37/page_protocol_basics.html

NET 网络结构体

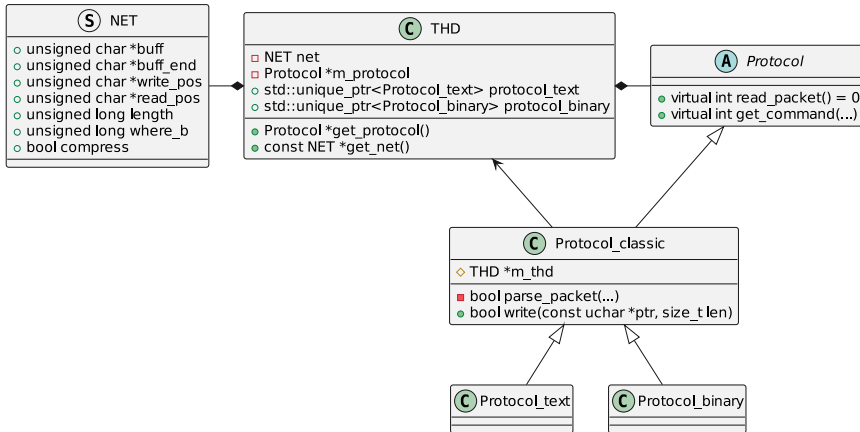
- vio 虚拟 IO 层的实现
- 一些缓冲的 buffer 定义
- 当前状态的指针：长度，Buffer 总长度，Buffer 所在位置
- 一些超时记录参数，读超时，写超时，重试超时
- compress 记录是否压缩

⑤ NET

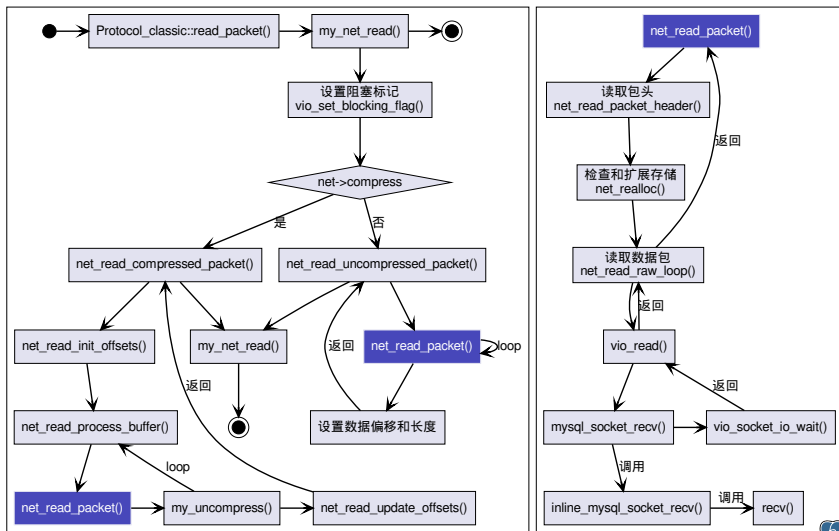
- MYSQL_VIO vio
- unsigned char *buff, *buff_end, *write_pos, *read_pos
- unsigned long remain_in_buf, length, buf_length, where_b
- unsigned long max_packet, max_packet_size
- unsigned int pkt_nr, compress_pkt_nr
- unsigned int write_timeout, read_timeout, retry_count
- int fcntl
- unsigned int *return_status
- unsigned char reading_or_writing
- unsigned char save_char
- bool compress
- unsigned int last_errno
- unsigned char error



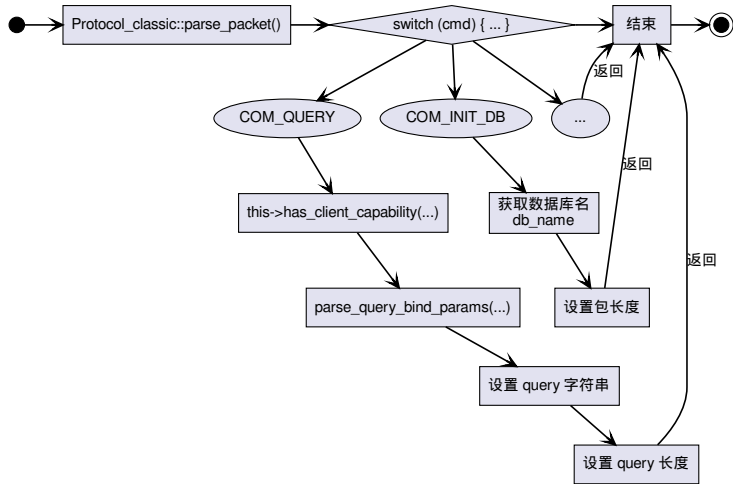
与 THD 之间的关系



读取包 read_packet 函数



解析包 parse_packet 函数



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派发模块

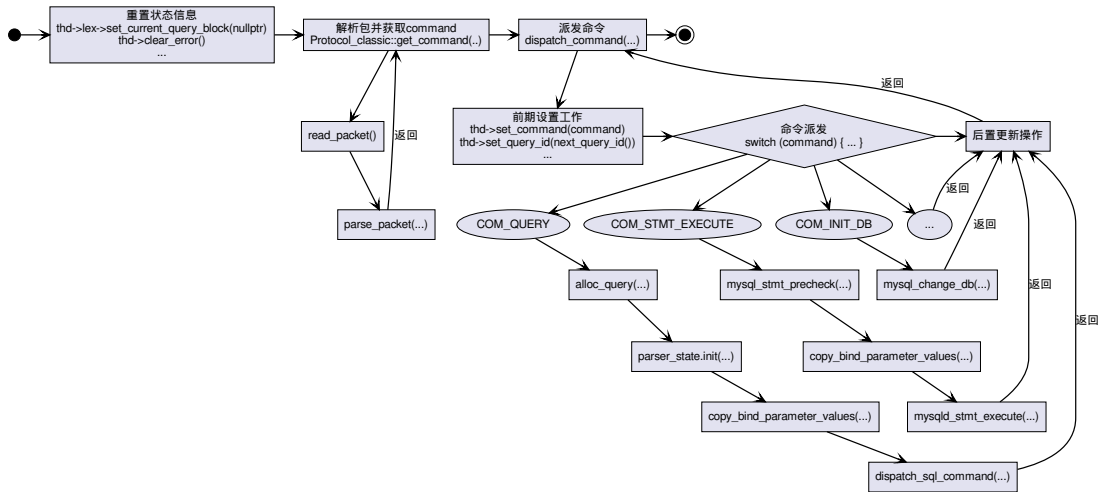


do_command 函数

- do_command() 的实际功能
 - 读取一个命令，注意命令可能是 query 或者指令
 - ① Query 例如: SELECT * FROM employees;
 - ② 指令例如: use dbname / shutdown
 - 然后执行命令，并发送结果
- do_command() 的实现见 [sql/sql_parse.cc](#)

```
1308 bool do_command(THD *thd) {
1309     bool return_value;
1310     int rc;
1311     :
1312     :
1313     :
1314     :
1315     :
1316     :
1317     :
1318     :
1319     :
1320     :
1321     :
1322     :
1323     :
1324     :
1325     :
1326     :
1327     :
1328     :
1329     :
1330     :
1331     :
1332     :
1333     :
1334     :
1335     :
1336     :
1337     :
1338     :
1339     :
1340     :
1341     :
1342     :
1343     :
1344     out:
1345     /* The statement instrumentation must be closed in all cases. */
1346     assert(thd->m_digest == nullptr);
1347     assert(thd->m_statement_psi == nullptr);
1348     return return_value;
1349 }
```

do_command 执行流程



结束

